

Recoverability-Limited Future Accessibility

A Minimal Computational Framework for Recoverability-Limited Future Accessibility

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1. Abstract

Many recovery-limited dynamical systems may remain dynamically active while progressively losing operational access to portions of their future accessibility landscape under repeated perturbation exposure.

Here we investigate whether such progressive accessibility loss may naturally generate structured operational temporal geometry under shared external temporal progression. Using a minimal family of recovery-limited dynamical systems subjected to repeated forcing, we observe the emergence of:

- i. divergent operational internal time accumulation,
- ii. operational temporal freezing,
- iii. operational temporal horizons,
- iv. probabilistic future accessibility under stochastic forcing,
- v. fragmentation of future-accessible regions accompanied by percolation-like connectivity degradation.

Importantly, these structures emerge despite identical external forcing schedules and identical externally elapsed time across systems.

Across the simulations, the central phenomenon is not immediate dynamical collapse, but progressive loss of operational future accessibility despite continued dynamical persistence. Under sufficiently constrained recoverability conditions, systems may remain operationally present while progressively losing access to portions of their future accessibility landscape.

The framework does not modify physical spacetime, relativistic metrics, or observer-dependent coordinate systems. Instead, the results suggest that recoverability-constrained systems may naturally generate heterogeneous operational temporal evolution and structured future accessibility geometry through internal recovery dynamics alone.

Within this interpretation, future operational accessibility becomes dependent not solely on externally elapsed time, but also on the evolving recoverability structure and accumulated recovery history of the system.

The present work is therefore intended as an exploratory computational framework for investigating recoverability-constrained future accessibility dynamics rather than as a physical theory of spacetime.

2. Introduction

2.1 Motivation

Many dynamical systems operate under finite recovery capacity while remaining continuously exposed to external perturbations. In such systems, the existence of a dynamically admissible

trajectory does not necessarily imply that the trajectory remains operationally reachable within finite recovery windows.

Traditional stability-oriented descriptions primarily characterize whether systems remain dynamically bounded or structurally persistent under forcing. However, recovery-limited systems may progressively lose future operational accessibility prior to complete structural collapse. Under these conditions, the relevant question becomes not only whether a future state exists, but whether the system retains sufficient recoverability to operationally access that future.

This distinction becomes particularly relevant in systems exhibiting:

- finite recovery rates,
- repeated perturbation exposure,
- delayed reorganization,
- heterogeneous internal timescales,
- and history-dependent recovery dynamics.

Examples of such behavior appear across biological regulation, adaptive systems, stress-response dynamics, distributed infrastructures, and non-equilibrium processes, where identical external forcing may generate substantially different future accessibility depending on the evolving internal recovery state.

Under this perspective, recoverability becomes associated not solely with static stability, but with the evolving accessibility structure of future operational trajectories.

2.2 Recoverability Perspective

Recent recoverability-oriented approaches have suggested that dynamical admissibility and operational accessibility may represent distinct properties in constrained systems. A trajectory may remain dynamically admissible within the underlying state space while simultaneously becoming operationally inaccessible due to finite recovery limitations.

Previous recoverability-based models have shown that systems subjected to repeated forcing may exhibit accessibility collapse prior to complete state-space collapse. These studies suggested that future operational evolution depends not only on perturbation magnitude, but also on the interaction between forcing duration, recovery dynamics, accumulated perturbation history, and delayed reorganization.

Operationally, the distinction may be summarized as:

admissibility $\neq$ recoverability

where admissibility refers to the existence of a dynamically allowed trajectory, whereas recoverability refers to the system's ability to operationally return to or access that trajectory under finite dynamical constraints.

Under repeated unresolved perturbations, accumulated recovery deficits may progressively contract the set of futures that remain operationally reachable. Consequently, future accessibility becomes history-dependent and partially constrained by the system's evolving recoverability structure.

The present work extends this perspective by investigating whether heterogeneous recoverability constraints alone may generate structured operational temporal accessibility under shared external temporal progression.

2.3 Operational Temporal Geometry

In recoverability-constrained systems, externally elapsed time alone may not fully characterize operational temporal evolution. Systems exposed to identical external temporal progression may nevertheless accumulate recoverable operational evolution at substantially different effective rates depending on their internal recovery dynamics.

To investigate this possibility, the present framework introduces the concept of operational internal time, defined through the accumulated recoverable dynamical activity sustained by the system under repeated forcing.

Within this operational interpretation:

- systems maintaining recoverability continue accumulating operational temporal progression,
- whereas systems approaching accessibility collapse exhibit progressively reduced operational temporal accumulation.

Under sufficiently constrained recovery conditions, systems may remain dynamically present while progressively losing operational access to portions of their future accessibility landscape.

This process may produce:

- operational temporal freezing,
- operational temporal horizons,
- and heterogeneous operational temporal trajectories despite identical external temporal progression.

Importantly, the framework does not interpret operational internal time as a physical clock or relativistic temporal coordinate. Rather, it represents an operational measure of accumulated recoverable evolution under finite recovery constraints.

The central question explored in the present work is therefore whether heterogeneous recoverability alone may naturally generate structured operational temporal accessibility under shared external temporal progression.

2.4 Scope and Limitations

The present framework is intentionally minimal and operational in scope. The model is not intended as a modification or replacement of relativistic spacetime theories, physical spacetime structure, or observer-dependent coordinate systems.

No relativistic metrics, Lorentz invariance, physical spacetime curvature, or physical propagation limits are introduced in the present formulation. Accordingly, terms such as:

- operational temporal accessibility,
- accessibility horizon,
- future accessibility cone,
- or temporal fragmentation

are used strictly in an operational dynamical sense within recoverability-limited systems.

Similarly, the term “observer” is used operationally to denote systems possessing distinct internal recovery dynamics under identical external forcing conditions, rather than conscious observers or relativistic reference frames.

The simulations employ intentionally minimal low-dimensional recovery-limited dynamics and should therefore be interpreted as exploratory computational probes rather than comprehensive descriptions of physical or biological systems.

The primary objective of the present work is to investigate whether heterogeneous recoverability constraints alone are sufficient to generate:

- divergent operational temporal evolution,
- structured future accessibility,
- operational temporal freezing,
- and operational temporal horizons

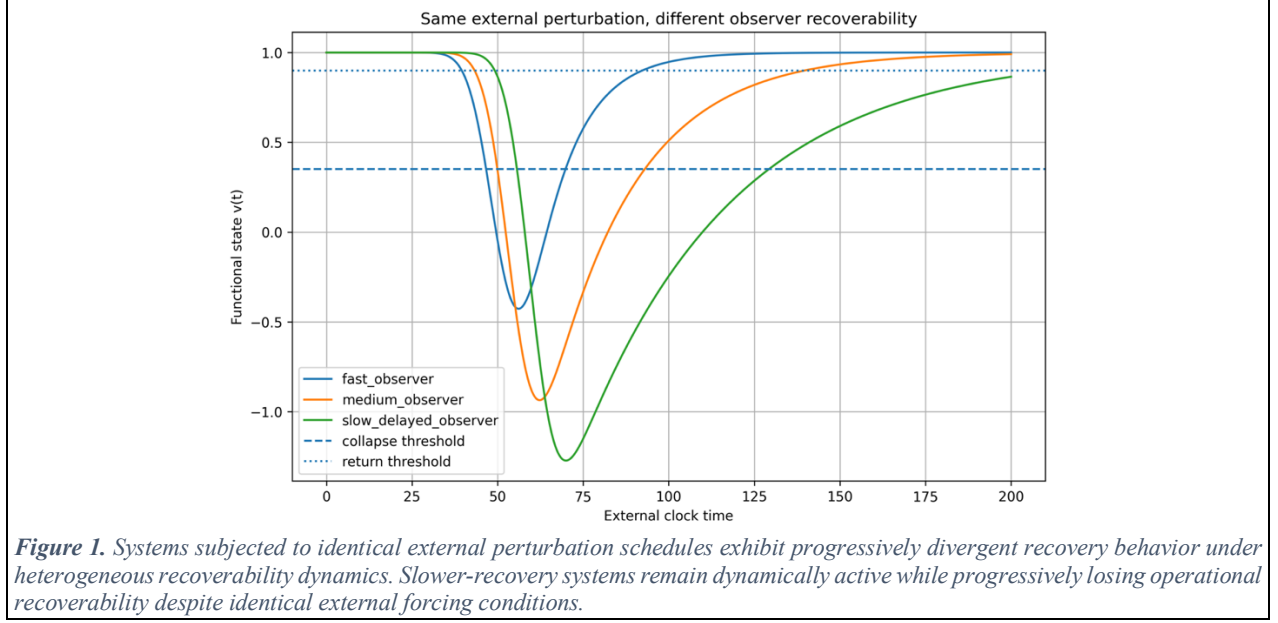
under shared external temporal progression.

Future work will be required to determine whether analogous accessibility structures emerge in higher-dimensional systems, empirical biological dynamics, or experimentally constrained physical systems.

3. Minimal Dynamical Framework

3.1 Recovery-Limited Dynamical System

To investigate recoverability-constrained operational temporal accessibility, we consider a minimal family of recovery-limited dynamical systems subjected to repeated external perturbations.



The operational state variable $v(t)$ evolves according to:

$$\frac{dv}{dt} = \frac{1 - v}{\tau_r} - E_0 u(t - \Delta)$$

where:

- $v(t)$ represents the instantaneous operational state of the system,
- τ_r denotes the intrinsic recovery timescale,
- Δ represents an optional recovery delay,
- E_0 controls perturbation amplitude,
- and $u(t)$ denotes external forcing.

The recovery term drives the system toward its nominal operational state, whereas the forcing term transiently reduces operational recoverability during perturbation exposure.

External perturbations are modeled as temporally localized Gaussian pulses:

$$u(t) = \frac{AUC}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(t - t_0)^2}{2\sigma^2}\right]$$

where:

- AUC controls total perturbation load,
- σ controls perturbation duration,
- and t_0 specifies perturbation timing.

Repeated perturbation sequences are generated through superposition of multiple pulses across external temporal progression.

Within this formulation, recoverability depends jointly on:

- perturbation load,
- forcing duration,
- recovery timescale,
- accumulated perturbation history,
- and delayed reorganization dynamics.

3.2 Heterogeneous Recoverability Systems

To investigate whether heterogeneous recoverability alone may generate divergent operational temporal evolution, we consider multiple systems evolving under identical external forcing conditions while possessing distinct internal recovery dynamics.

Three representative recovery regimes are examined throughout the simulations:

- fast recovery,
- intermediate recovery,
- and slow recovery.

All systems share:

- identical perturbation schedules,
- identical externally elapsed duration,
- and identical numerical integration procedures.

However, systems differ in their internal recovery parameters, including:

- recovery timescale τ_r ,
- recovery delay Δ ,
- and operational recovery-window structure.

Throughout the present work, the term “observer” is used operationally to denote systems possessing distinct internal recoverability dynamics under common external forcing conditions. No observer-dependent coordinate transformations or relativistic reference frames are introduced.

Under this framework, systems evolve under identical external temporal progression while potentially accumulating substantially different operational temporal evolution due solely to heterogeneous recoverability structure.

This formulation allows direct investigation of whether:

- future accessibility contraction,
- operational temporal freezing,
- and accessibility horizons

may emerge from heterogeneous recovery dynamics alone.

3.3 Future Accessibility Probes

Future operational accessibility is evaluated through prospective perturbation probes applied at successive stages of system evolution.

At a given external time t_c , the instantaneous system state $\mathbf{v}(t_c)$ is extracted and subjected to an additional perturbation probe characterized by:

- fixed perturbation duration,
- controlled perturbation load,
- and finite recovery observation window.

The objective of the probe is to determine whether the system retains sufficient recoverability to operationally return to an accessible state following additional future forcing.

Operational future accessibility is defined conditionally:

- a future region is considered operationally accessible if recoverability is successfully restored within the observation window,
- whereas a future region is considered inaccessible if recoverability fails within that finite interval.

Repeated probing across external temporal progression and perturbation-load parameter space generates evolving future accessibility landscapes whose structure depends on:

- accumulated recovery deficits,
- perturbation history,
- forcing duration,
- delayed recovery,
- and heterogeneous recoverability dynamics.

Importantly, these accessibility structures emerge despite identical external forcing schedules and identical externally elapsed temporal progression across systems.

Within this interpretation, future accessibility becomes an evolving property of the system's recoverability state rather than a static consequence of externally elapsed time alone.

Repeated future probing produces structured accessibility phase diagrams allowing identification of:

- recoverable future regions,
- inaccessible future regions,
- accessibility boundaries,
- and operational accessibility transitions.

4. Operational Internal Time

4.1 Operational Internal Time Accumulation

Under shared external temporal progression, systems possessing distinct recoverability constraints may nevertheless accumulate operational evolution at substantially different effective rates.

To characterize this behavior, the framework introduces an operational temporal accumulation variable $\tau_{int}(t)$ defined through accumulated recoverable dynamical activity over externally elapsed time:

$$\tau_{int}(t) = \int_0^t \max(v(t'), 0), dt'$$

where $v(t)$ denotes the instantaneous operational state of the system.
Within this formulation:

- systems maintaining recoverability continue accumulating operational temporal progression,
- whereas systems approaching accessibility collapse exhibit progressively reduced operational temporal accumulation

Importantly, $\tau_{int}(t)$ is not interpreted as a physical clock or relativistic temporal coordinate. Rather, it represents an operational measure of accumulated recoverable evolution under finite recovery constraints.

Under heterogeneous recoverability conditions, systems subjected to identical external temporal progression may therefore accumulate substantially different operational temporal evolution despite sharing identical externally elapsed duration.

This formulation allows direct investigation of whether divergent operational temporal evolution may emerge naturally from recoverability-limited dynamics alone.

4.2 Operational Temporal Divergence

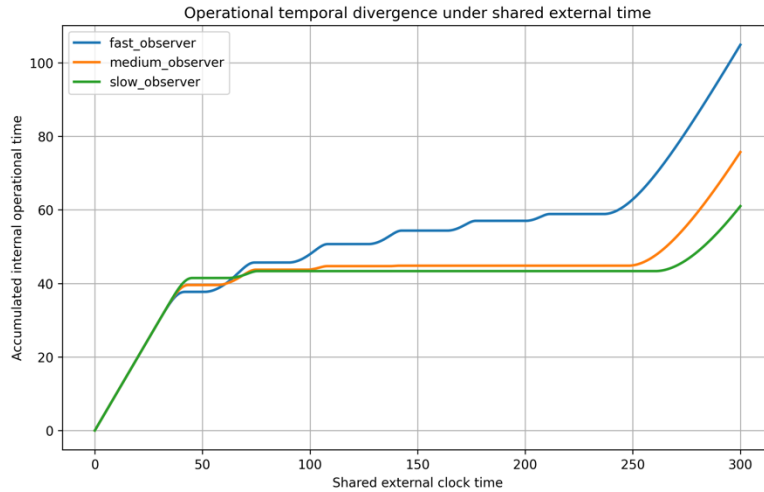


Figure 2. Under identical external temporal progression, systems possessing slower recoverability progressively accumulate reduced operational evolution while remaining dynamically present. Constrained recoverability progressively limits future operational accessibility despite identical perturbation schedules and externally elapsed time.

Simulations performed under identical external temporal progression reveal substantial divergence in operational temporal accumulation across systems possessing distinct recoverability structure.

Fast-recovery systems generally maintain continuous operational temporal accumulation throughout repeated perturbation exposure. In contrast, slower-recovery systems progressively exhibit reduced operational accumulation following unresolved perturbation sequences and delayed reorganization.

Under sufficiently constrained recovery conditions, operational temporal accumulation may partially stall over extended intervals despite continued external temporal progression. Consequently, systems may remain dynamically present while progressively losing operational access to portions of their future accessibility landscape.

Importantly, this divergence emerges solely from heterogeneous recovery dynamics and does not require:

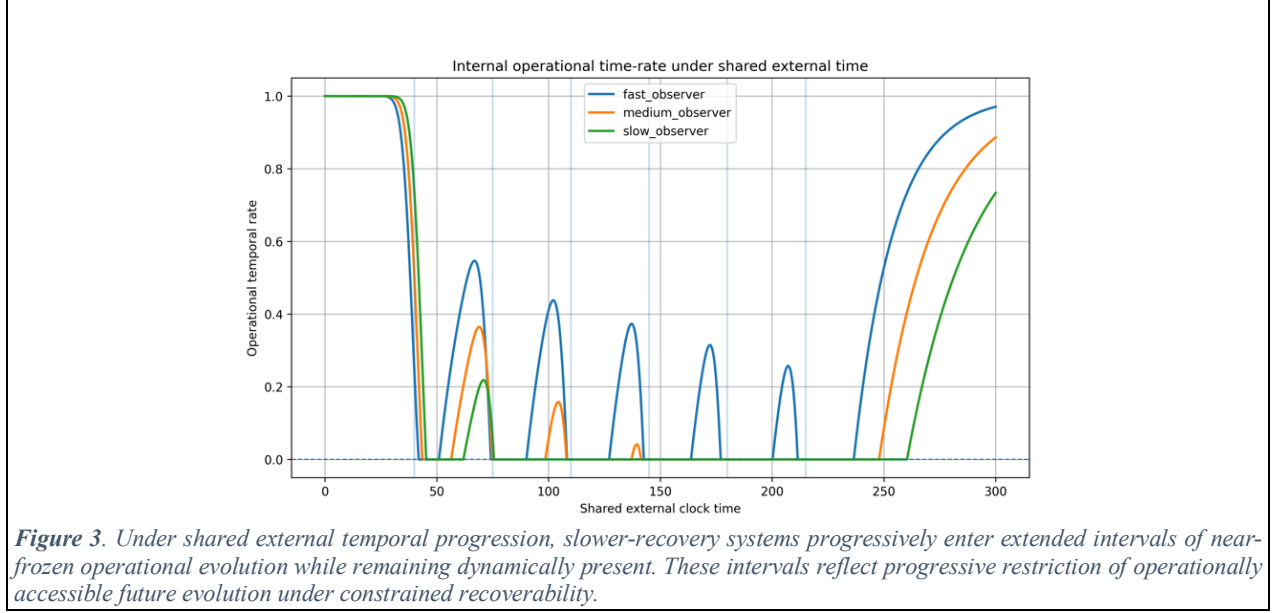
- differing external forcing,
- differing physical clocks,
- relativistic temporal effects,
- or observer-dependent coordinate transformations.

Instead, the simulations suggest that heterogeneous recoverability alone may generate substantially different operational temporal trajectories under identical external temporal progression.

Operational temporal divergence is also strongly history-dependent. Accumulated unresolved perturbations progressively constrain future recoverability, thereby reducing subsequent operational temporal accumulation and contracting future operational accessibility.

Within this interpretation, operational temporal evolution becomes partially constrained by the evolving recoverability state of the system rather than solely by externally elapsed time.

4.3 Operational Temporal Freezing and Horizons



To characterize intervals of severely reduced operational evolution, the framework evaluates the operational temporal accumulation rate:

$$\frac{d\tau_{int}}{dt}$$

which measures the instantaneous rate of operational temporal accumulation relative to external temporal progression.

Under repeated forcing, slower-recovery systems frequently exhibit extended intervals during which:

$$\frac{d\tau_{int}}{dt} \approx 0$$

despite continued external temporal evolution.

These intervals correspond to regimes of operational temporal freezing in which the system remains dynamically active while accumulating little or no additional recoverable operational evolution.

To characterize these regimes operationally, the framework defines operational horizons as extended intervals of near-frozen operational temporal accumulation arising under constrained recoverability conditions.

Importantly, these horizons do not represent physical spacetime boundaries or relativistic event horizons. Rather, they constitute operational accessibility limitations emerging from recoverability-constrained dynamical evolution.

Simulations reveal that horizon persistence scales strongly with recoverability structure:

- fast-recovery systems typically exhibit short-lived freezing intervals followed by renewed operational accumulation,
- whereas slower-recovery systems may enter prolonged or persistent operational horizon regimes under repeated unresolved forcing.

Under sufficiently constrained recovery conditions, systems may therefore preserve dynamical persistence while progressively losing operational access to portions of their future accessibility landscape.

This process generates progressively constrained future accessibility structure under shared external temporal progression.

5. Evolution of Future Accessibility Structure

5.1 Future Accessibility Landscapes

Repeated future accessibility probing generates dynamically evolving operational accessibility landscapes across external temporal progression and perturbation-load parameter space.

At successive external time points, systems are subjected to prospective perturbation probes evaluating whether recoverability remains operationally achievable following additional future forcing.

The resulting accessibility landscapes reveal that future operational reachability is not uniformly preserved throughout system evolution. Instead, recoverable and inaccessible future regions emerge progressively as accumulated recovery deficits alter the evolving accessibility structure.

Importantly, accessibility boundaries evolve continuously rather than appearing as instantaneous collapse transitions. Under increasing recoverability constraints, future-accessible regions may:

- contract,
- become increasingly constrained,
- or become partially disconnected

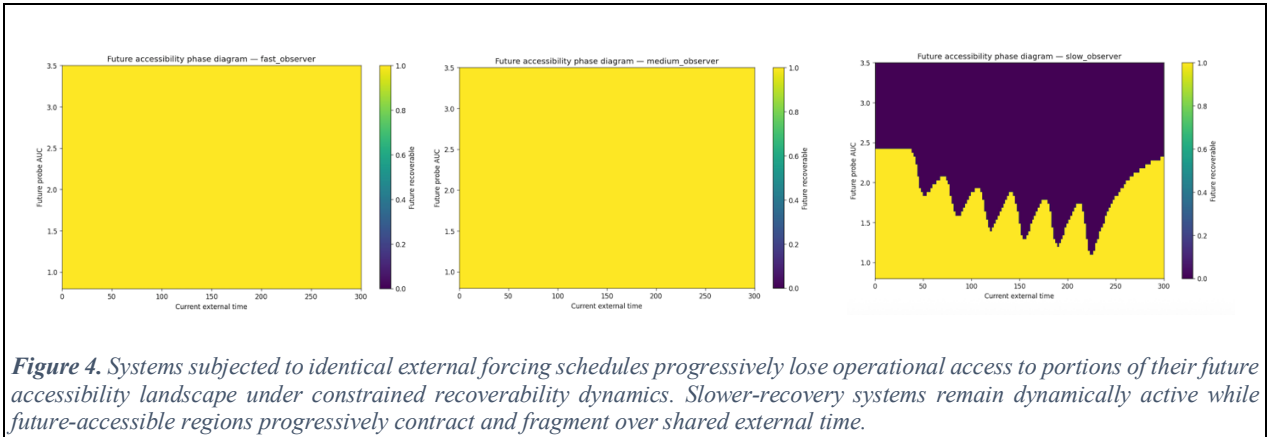
despite continued dynamical persistence.

Within this interpretation, future accessibility becomes an evolving property of the system's recoverability state rather than a static consequence of externally elapsed time alone.

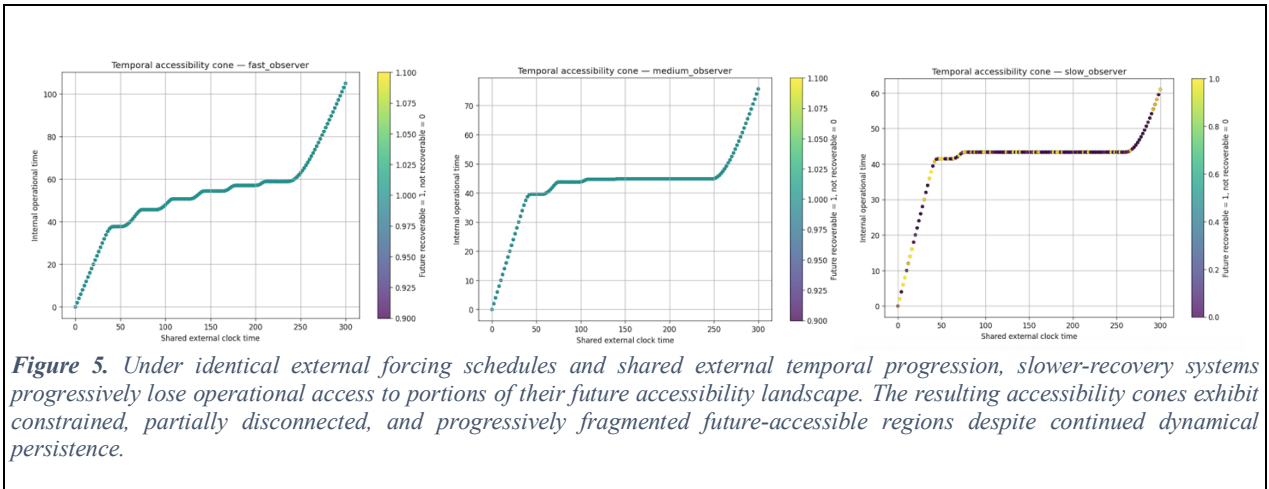
The resulting accessibility landscapes exhibit:

- structured recoverable regions,
- inaccessible future sectors,
- evolving accessibility boundaries,
- and operational accessibility transitions.

These structures emerge despite identical external forcing schedules and identical external temporal progression across systems.



5.2 Temporal Accessibility Cones



To visualize the relationship between operational temporal evolution and future recoverability, the framework combines:

- external temporal progression,
- accumulated operational temporal accumulation,

- and prospective future accessibility

within a unified accessibility representation.

For each system, future accessibility probes are evaluated continuously throughout dynamical evolution while simultaneously tracking accumulated operational temporal accumulation. This procedure generates structured accessibility regions in the space defined by:

- external temporal progression,
- operational temporal accumulation,
- and future recoverability outcome.

The resulting structures form evolving operational accessibility cones in which future operational reachability progressively depends on the system's accumulated recoverability history.

Importantly, the term “cone” is used purely as an operational accessibility descriptor and does not imply relativistic causal geometry or physical light-cone structure.

Rather, these structures constitute operational accessibility geometries emerging from recoverability-constrained dynamics.

Fast-recovery systems typically preserve broad continuously accessible future regions throughout repeated perturbation exposure. In contrast, slower-recovery systems progressively develop:

- contracted accessibility regions,
- persistent operational inaccessibility regimes,
- partially disconnected future accessibility sectors,
- and extended inaccessible future regions associated with operational temporal freezing.

Under sufficiently constrained recoverability conditions, accessibility cones may partially collapse or become progressively disconnected despite continued external temporal progression.

Within this interpretation, future operational accessibility becomes increasingly constrained by accumulated recovery history rather than solely by externally elapsed time.

These results suggest that recoverability-constrained systems may naturally generate structured operational accessibility geometry through internal recovery dynamics alone.

6. Stochastic Accessibility Fragmentation

6.1 Stochastic Future Accessibility

To evaluate robustness under perturbation variability, future accessibility probing was extended to include additive stochastic forcing during prospective recovery simulations.

Under deterministic conditions, future accessibility boundaries remain relatively sharp, separating recoverable and inaccessible future regions across parameter space. Under stochastic forcing, however, these boundaries progressively transform into probabilistic accessibility landscapes.

In this regime, future operational accessibility is no longer strictly binary. Instead, identical systems subjected to equivalent average forcing conditions may exhibit probabilistic recoverability outcomes depending on stochastic perturbation realization.

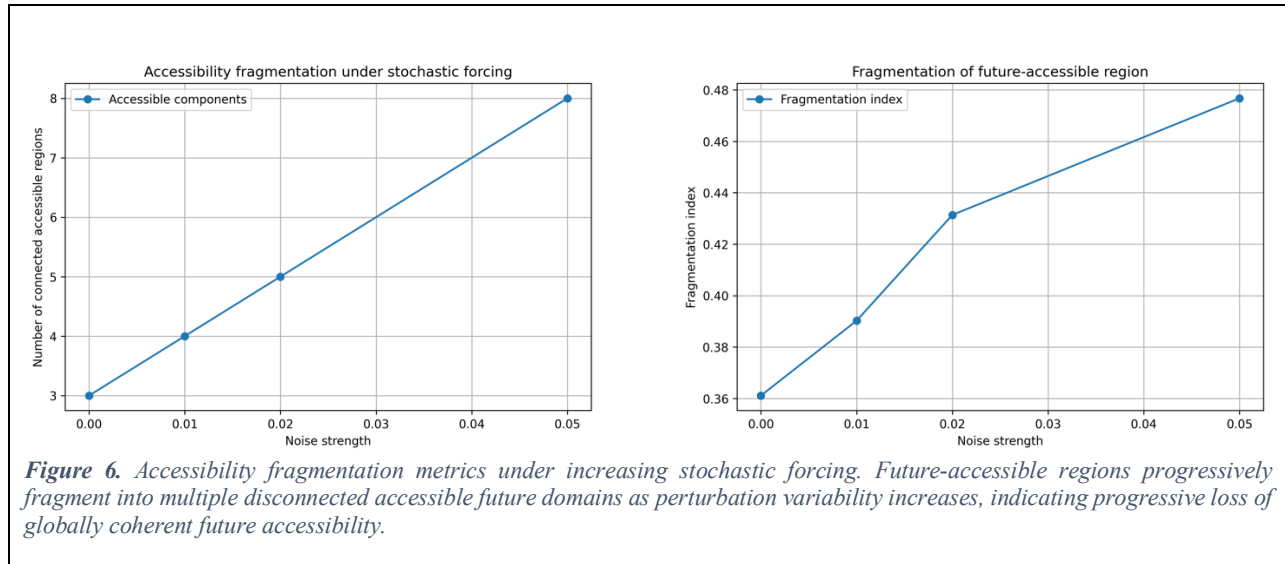
Importantly, the resulting accessibility structure remains spatially organized rather than collapsing into unstructured noise. Accessibility transitions continue to emerge preferentially within specific regions of the evolving accessibility landscape, indicating that the underlying recoverability geometry persists under stochastic forcing.

Under increasing stochasticity, the framework exhibits:

- probabilistic future accessibility,
- broadened accessibility transitions,
- partial accessibility persistence,
- and structured stochastic accessibility geometry.

These results suggest that future operational accessibility may remain geometrically organized even when deterministic recoverability boundaries become probabilistic.

6.2 Accessibility Fragmentation and Percolation



Under increasing stochastic forcing, future-accessible regions progressively fragment into multiple disconnected operational accessibility components.

To characterize this behavior, future accessibility landscapes were analyzed using connectivity-based metrics applied to recoverable future regions across perturbation-load and temporal parameter space.

Under low stochastic forcing, future-accessible regions typically remain relatively coherent, forming large connected operational accessibility domains. However, increasing stochastic variability progressively transforms these coherent regions into fragmented accessibility structures composed of smaller disconnected operational sectors.

Importantly, stochastic forcing does not simply eliminate future accessibility globally. Instead, the simulations reveal a transition toward locally preserved but globally fragmented future operational accessibility.

This fragmentation process is accompanied by:

- increasing numbers of disconnected accessibility components,
- reduced global accessibility coherence,
- declining largest-component dominance,
- and percolation-like degradation of future operational connectivity.

Consequently, systems may preserve isolated pockets of recoverability while progressively losing globally connected future accessibility.

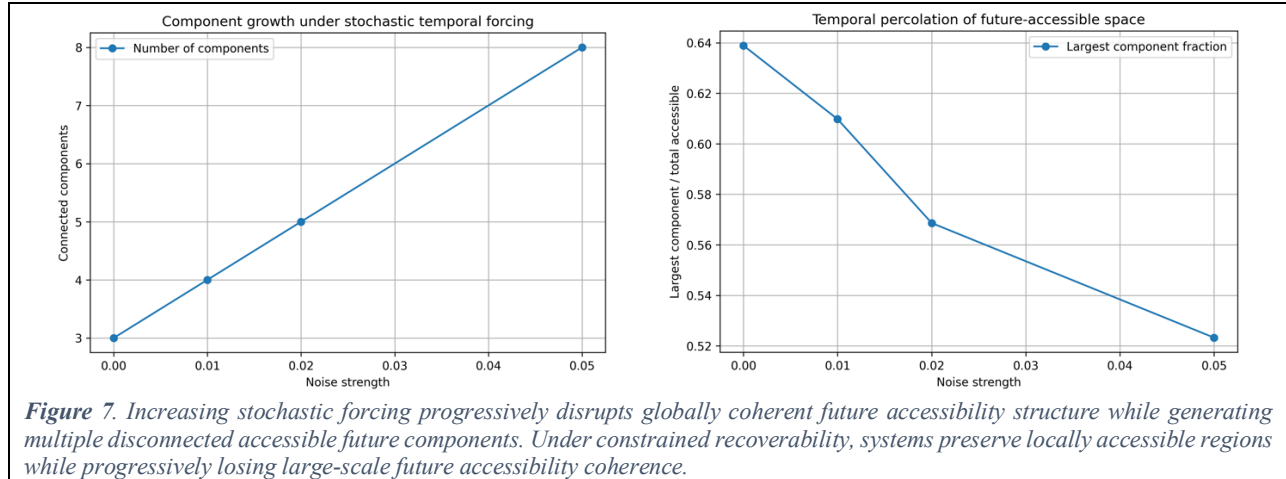
The resulting accessibility geometry exhibits characteristics analogous to percolation-like transitions in which future operational reachability progressively loses large-scale coherence under constrained recoverability and stochastic forcing.

Importantly, these fragmented accessibility structures emerge despite:

- identical external forcing schedules,
- identical external temporal progression,
- and unchanged underlying state-space dimensionality.

The observed fragmentation therefore appears to emerge primarily from recoverability-constrained accessibility dynamics rather than from explicit restructuring of the underlying dynamical topology itself.

Within this interpretation, future operational accessibility becomes not only probabilistic but also geometrically fragmented under constrained recoverability conditions.



To further characterize fragmentation dynamics, the framework evaluates the evolution of global accessibility connectivity under increasing stochastic forcing.

Connectivity analysis reveals that stochastic forcing progressively reduces the coherence of globally connected future-accessible regions while simultaneously increasing fragmentation into smaller disconnected operational sectors.

In particular, increasing stochastic variability produces:

- declining largest-component fraction,
- reduced global connectivity coherence,
- increasing accessibility component proliferation,
- and progressive degradation of large-scale operational accessibility continuity.

These results suggest that future operational accessibility may undergo percolation-like degradation in which globally coherent accessibility structure progressively collapses into locally preserved but disconnected future-accessible regions.

Importantly, the underlying state-space dimensionality remains unchanged throughout the simulations. The observed connectivity degradation therefore emerges operationally through constrained recoverability dynamics rather than through explicit topological restructuring of the dynamical system itself.

To evaluate the structural organization of future-accessible trajectories, recoverable accessibility regions were additionally analyzed using trajectory-network connectivity metrics.

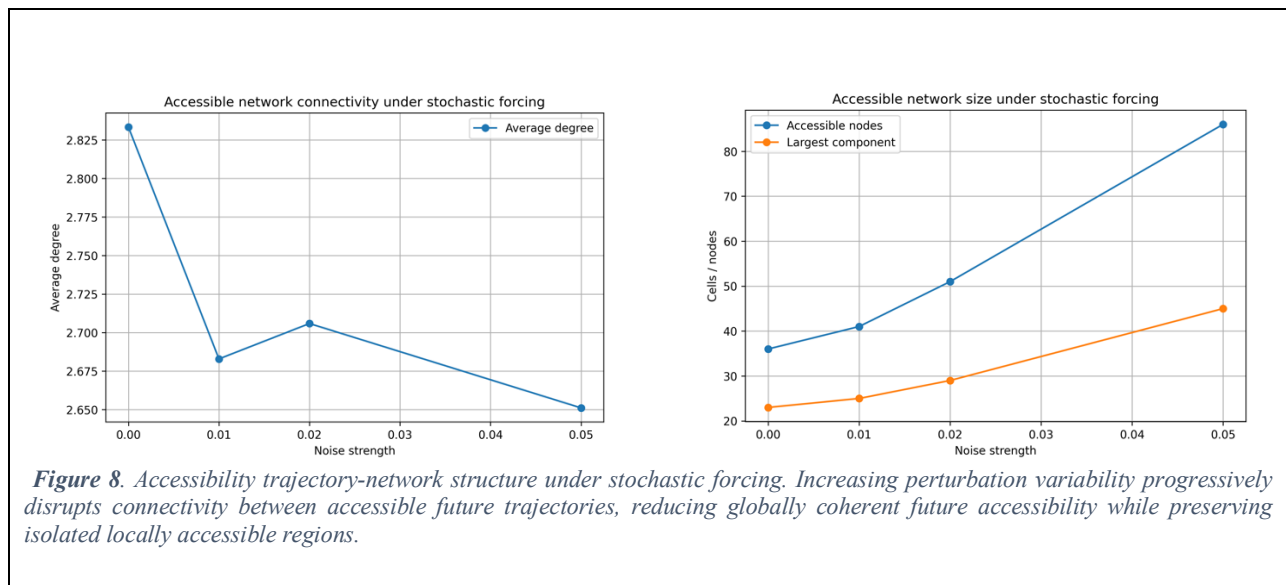
Within this representation, recoverable operational states form evolving accessibility networks whose structure depends on recoverability persistence across perturbation and temporal parameter space.

Under increasing stochastic forcing, these trajectory networks progressively exhibit:

- reduced average connectivity,
- increasing network sparsification,
- declining accessibility coherence,
- and fragmentation into partially disconnected operational accessibility subnetworks.

Despite this degradation, portions of the accessibility network frequently remain locally preserved even under substantial stochastic forcing, indicating partial persistence of operational recoverability structure.

These results further support the interpretation that recoverability-constrained systems may naturally generate fragmented operational accessibility geometry under stochastic perturbation exposure.



7. Homogeneous Controls

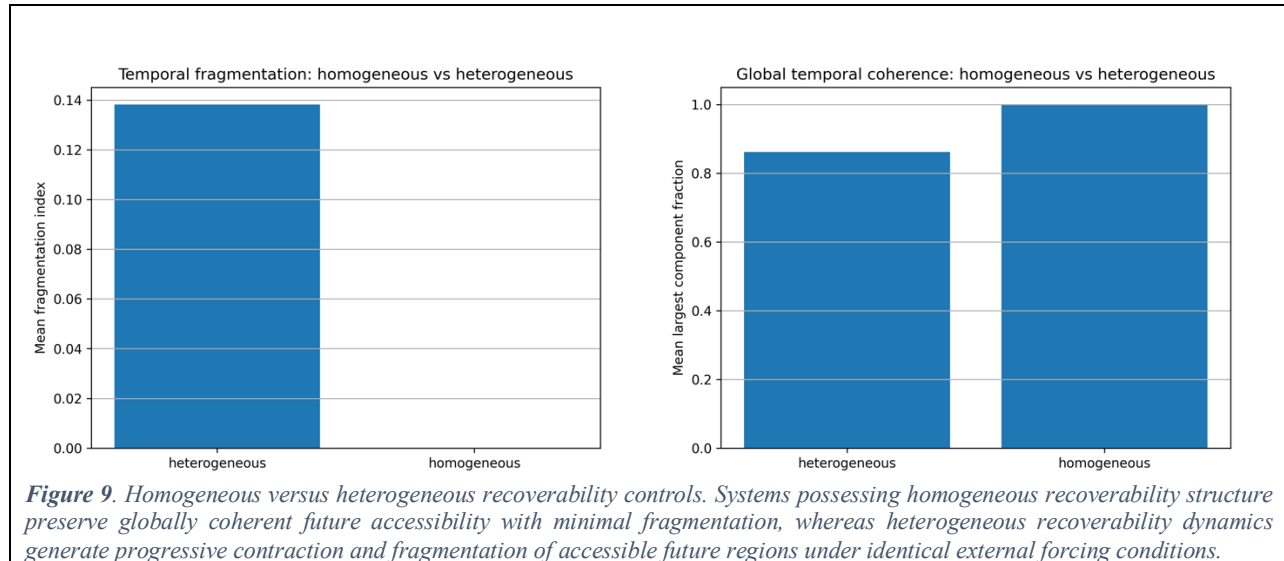
7.1 Homogeneous Recoverability Controls

To evaluate whether the observed accessibility fragmentation emerges specifically from heterogeneous recoverability structure, additional control simulations were performed using homogeneous recovery dynamics.

In these control systems, all simulated systems shared:

- identical recovery timescales,
- identical recovery delays,
- identical operational recovery-window structure,
- and identical external forcing conditions.

Under homogeneous recoverability conditions, systems exhibited substantially reduced operational temporal divergence compared to heterogeneous simulations. Future accessibility landscapes remained significantly more coherent, with minimal fragmentation and greatly reduced formation of disconnected accessibility regions.



Importantly, stochastic forcing alone did not reproduce the degree of fragmentation observed in heterogeneous recoverability systems.

Connectivity analysis revealed that homogeneous systems preserved:

- larger connected accessibility domains,
- stronger global accessibility coherence,
- reduced fragmentation indices,
- and substantially lower accessibility component proliferation under stochastic forcing.

In contrast, heterogeneous recoverability systems exhibited:

- persistent operational temporal divergence,
- contracted future accessibility,
- extended operational freezing intervals,
- and progressive fragmentation of globally connected future accessibility.

These results suggest that the observed accessibility geometry does not emerge solely from stochastic perturbation variability or repeated forcing exposure.

Instead, heterogeneous internal recoverability structure appears to play a central role in generating fragmented future operational accessibility under shared external temporal progression.

The control simulations therefore support the interpretation that heterogeneous recovery dynamics themselves contribute directly to the emergence of:

- operational temporal divergence,
- fragmented accessibility geometry,
- horizon-like accessibility structure,
- and progressive loss of globally coherent future recoverability.

Within this interpretation, recoverability heterogeneity acts not merely as a quantitative modifier of recovery efficiency, but as a structural determinant of future operational accessibility organization.

8. Discussion

8.1 Recoverability-Constrained Future Accessibility

The central phenomenon observed throughout the simulations is not immediate dynamical collapse, but progressive loss of operational future accessibility despite continued dynamical persistence.

Under constrained recoverability conditions, systems may remain dynamically active while progressively losing operational access to portions of their future accessibility landscape. Within this interpretation, operational temporal geometry emerges as a secondary structural consequence of evolving accessibility loss rather than as the primary phenomenon itself.

The present framework suggests that recoverability-constrained systems evolving under identical external temporal progression may nevertheless develop substantially different operational temporal trajectories depending on their internal recovery dynamics.

Importantly, the observed divergence does not emerge from differing physical clocks, relativistic observer frames, or modifications of physical spacetime structure. Instead, the simulations indicate that heterogeneous recoverability alone may progressively constrain future operational accessibility under repeated forcing.

Several operational structures consistently emerged throughout the simulations:

- divergent operational temporal accumulation,
- operational temporal freezing,
- horizon-like accessibility regimes,
- fragmented future accessibility,
- and percolation-like degradation of global accessibility coherence under stochastic forcing.

These structures emerged despite:

- identical external forcing conditions,
- identical externally elapsed duration,
- and unchanged underlying state-space dimensionality.

Within this interpretation, future operational accessibility becomes a dynamically evolving property of the system's recoverability state rather than a static consequence of externally elapsed time alone.

Importantly, the framework does not imply that physical time itself evolves differently across systems. Rather, the results suggest that systems possessing distinct recoverability structure may accumulate recoverable operational evolution at substantially different effective rates under identical external temporal progression.

Under this perspective, operational temporal divergence emerges naturally from constrained recoverability dynamics rather than from observer-dependent physical temporal structure.

More broadly, the simulations suggest that future operational accessibility may progressively contract, fragment, or become operationally disconnected under constrained recoverability conditions despite continued dynamical persistence.

Within this interpretation, the observed operational temporal geometry emerges from the evolving structure of accessible futures rather than from modifications of externally elapsed time itself.

8.2 Relation to Existing Dynamical Frameworks

The operational structures observed in the present framework exhibit conceptual relationships with several established areas of dynamical systems research, including:

- recoverability-constrained dynamics,
- rate-induced transitions,
- stochastic accessibility,
- critical transitions,
- network fragmentation,
- and percolation-like connectivity degradation.

Previous studies in recoverability-limited and rate-induced dynamical systems have demonstrated that systems may lose return accessibility despite remaining dynamically bounded. Similarly, stochastic dynamical systems and critical-transition frameworks have shown that structured probabilistic transitions may emerge under increasing perturbation variability.

The present framework extends these perspectives by explicitly examining how heterogeneous recoverability constraints may generate evolving operational future accessibility geometry under common external temporal progression.

In particular, the observed transition from:

- coherent accessibility,
- to probabilistic accessibility,
- to fragmented accessibility structure

resembles connectivity degradation and percolation-like transitions observed in non-equilibrium and networked systems.

However, the present framework differs from classical fragmentation approaches in an important respect: the underlying state-space topology itself is not explicitly restructured. Instead, fragmentation emerges operationally through constrained recoverability acting on future accessibility.

Similarly, the framework shares conceptual similarities with accessibility landscapes explored in control theory and dynamical reachability analysis, where future evolution depends on evolving system constraints.

The present formulation extends these perspectives by incorporating:

- repeated forcing,
- finite recoverability,
- delayed reorganization,
- heterogeneous recovery dynamics,
- and operational temporal accumulation.

Importantly, the framework remains intentionally minimal and exploratory. The present results do not establish physical temporal laws or relativistic causal structure.

Rather, the results suggest that recoverability-limited systems may progressively lose operational access to future regions long before complete dynamical collapse occurs.

Under sufficiently constrained conditions, this evolving accessibility loss naturally generates structured operational temporal geometry, fragmented future accessibility, and horizon-like operational regimes.

8.3 Limitations and Future Directions

The present framework is intentionally minimal and exploratory in scope. Several important limitations should therefore be emphasized.

First, the simulations employ low-dimensional recovery-limited dynamical systems designed to investigate operational accessibility structure rather than to reproduce specific physical, biological, or engineered systems in detail.

Consequently, the present results should be interpreted as computational demonstrations of possible recoverability-driven accessibility phenomena rather than direct descriptions of empirical systems.

Second, the framework does not incorporate:

- relativistic dynamics,
- physical spacetime geometry,
- Lorentz invariance,
- physical causal propagation limits,
- or observer-dependent coordinate systems.

Accordingly, the operational temporal structures described in the present work should not be interpreted as modifications of physical time or relativistic spacetime structure.

Third, the current simulations investigate relatively simple forms of:

- recoverability limitation,
- stochastic forcing,
- delayed recovery,
- and repeated perturbation exposure.

Higher-dimensional systems, adaptive recovery dynamics, networked substrates, or experimentally constrained systems may exhibit substantially more complex accessibility behavior.

Future work will therefore be required to determine:

- whether analogous accessibility structures emerge in empirical biological systems,
- whether operational fragmentation persists in higher-dimensional dynamics,
- whether recoverability-constrained accessibility geometry generalizes across broader dynamical classes,
- and whether operational temporal accumulation provides predictive structure in experimentally measurable systems.

Potential applications include:

- biological stress-response systems,
- hypoxia and recovery dynamics,
- distributed adaptive systems,
- resilience analysis,
- and recoverability-constrained control frameworks.

The present work should therefore be interpreted as an initial exploratory computational framework investigating whether heterogeneous recoverability constraints alone may generate structured operational temporal geometry under shared external temporal progression.

9. Conclusion

The present work investigated whether heterogeneous recoverability constraints alone may generate structured operational temporal accessibility geometry under shared external temporal progression.

Using a minimal family of recovery-limited dynamical systems subjected to repeated perturbation exposure, the simulations consistently exhibited:

- divergent operational temporal accumulation,
- operational temporal freezing,
- horizon-like accessibility regimes,
- probabilistic future accessibility under stochastic forcing,
- and fragmentation of future-accessible regions accompanied by percolation-like degradation of global accessibility coherence.

Importantly, these structures emerged despite identical external forcing schedules, identical externally elapsed duration, and unchanged underlying state-space dimensionality across systems.

Within the present framework, systems may remain dynamically active while progressively losing operational access to portions of their future accessibility landscape. Under sufficiently constrained recoverability conditions, future operational accessibility becomes increasingly dependent not solely on externally elapsed time, but also on the evolving recoverability structure and accumulated recovery history of the system.

The observed operational temporal divergence does not arise from differing physical clocks, relativistic observer frames, or modifications of physical spacetime structure. Rather, the simulations suggest that heterogeneous internal recoverability alone may naturally generate substantially different operational accessibility trajectories under identical external temporal progression.

Similarly, the emergence of fragmented future accessibility under stochastic forcing suggests that recoverability-constrained systems may preserve locally accessible operational regions while progressively losing globally coherent future accessibility structure.

More broadly, the present results suggest that constrained dynamical systems may progressively lose operational access to future regions long before complete dynamical collapse occurs.

Within this interpretation, future accessibility becomes an evolving recoverability-dependent property rather than a static consequence of externally elapsed time alone. Structured operational temporal geometry therefore emerges not as a primary assumption, but as an emergent organizational consequence of constrained future accessibility dynamics.

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